SERIES G

A Phenomenological Vacuum-Lattice Framework:

Mapping Six Phenomena to a Single Geometric Offset $v^2 = 0.00017$

lambda = 1.159e-97 (λ/4 PRD standard) · f_c = 18.52 nHz · xi_0 = 0.2722 ly · 1.2σ seed · p=0.231 · Tier A/B falsification

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Target: Foundations of Physics or Physical Review D (Perspective)

Series A: doi:10.5281/zenodo.19443940 | Series B: doi:10.5281/zenodo.19444436 | Series D: doi:10.5281/zenodo.19446649 | Series G of IX

SERIES G CORRECTIONS (Omega audit): (1) DARK MATTER VORTEX SPACING: n = m_phi v_flat²/hbar = 5.17×10■¹■ m■², giving d = 1/sqrt(n) = 4398 km (not 14,000 km as originally stated). The paper's n is correct; the d calculation was off by factor ~3. Corrected to ~4,400 km. The paper already caveats this section heavily. (2) SONOLUMINESCENCE COLOR: n=5 gives 477 THz = 629 nm = orange-red (not blue-green). Corrected to 'visible orange-red.' The prediction for visible light remains valid. (3) All five sync items from directive applied: λ/4 notation, 1.2σ/p=0.231, f_c/m_phi/xi anchors, lab scaling relations exact, Appendix C quarantined.

ABSTRACT

We explore a phenomenological framework in which the cosmological vacuum is assigned a single geometric torsion offset $v^2 = 0.00017$. This offset provides a common scale mappable onto several disparate phenomena: a candidate 1.73 Mpc clustering excess in JWST high-redshift galaxies (global significance 1.20σ after look-elsewhere correction, p = 0.231, Paper E); a narrow 18.52 nHz feature in the nanohertz gravitational-wave background; and a predicted sidereal modulation of the forbidden reflection in nanodiamonds. We present a minimal effective Lagrangian in the λ/4 normalization (PRD standard): $V = (\lambda/4)(\phi^2 - v^2\Lambda^2)^2$, with λ = 1.16×10■■■ (technically natural; related to Paper D's convention by λ_G = 4×λ_D). Derived scales: m_phi = 7.660e-23 eV, xi_0 = 0.2722 ly, f_c = 18.52 nHz. In the limit $v^2 \to 0$, the lattice disappears and the framework reduces to ΛCDM + QFT. Laboratory predictions include: a 170 ppm directional Casimir torque ($v^2 \times \sin(2\theta)$), a quantum-eraser relaxation time tau_lab = 1.23e-04 s, and a sonoluminescence harmonic at n=5 → 477 THz (629 nm, visible orange-red). All speculative extensions (warp-drive analogues, metamaterials, tachyonic phase waves) are confined to Appendix C with explicit 'inspirational only' disclaimers. The framework is falsifiable through five distinct experimental channels.

"The universe is not an explosion; it is a tuned, resonating crystal. The snowflake in the lab is the same as the lattice in the deep field."

FIVE SYNCHRONIZATIONS (Directive Lock)

1. λ/4 notation: $V=(\lambda/4)(\phi^2-v^2\Lambda^2)^2 \to \lambda = 1.159e-97$ (= 4×λ_D)

2. Honest Shield: 1.20σ global, p=0.231 — 'tentative seed, not detection'

3. Anchors locked: f_c=18.52 nHz, m_phi=7.66e-23 eV, xi=0.2722 ly

4. Lab scaling exact: Casimir 170 ppm, tau_lab=1.23e-04 s, sono n=5→629 nm

5. Appendix C quarantined: warp-drive, metamaterials, tachyon — 'inspirational only'

1. Introduction: The Power of One Parameter

Standard quantum field theory treats the vacuum as a featureless background. The MSDS programme (Papers A-F) has proposed that the vacuum instead has a geometric torsion offset $v^2 = 0.00017$ — a single, small, testable constant. Paper G maps this constant onto six disparate phenomena, demonstrating the explanatory reach of the framework and motivating targeted experiments. Full quantitative derivations for each phenomenon are deferred to companion papers.

Two important caveats govern this paper. First, the 1.73 Mpc clustering excess (Paper E) has a global significance of only 1.20σ (p = 0.231). This feature is not strong enough to claim a detection; it is a tentative seed worth further investigation. Second, several competing explanations exist for each phenomenon: primordial non-Gaussianity, scale-dependent bias, survey systematics. MSDS proposes a vacuum-lattice candidate that makes distinct, testable predictions. The data will decide.

Limiting case: In the limit $v^2 \to 0$, the lattice disappears and the framework reduces identically to standard ΛCDM cosmology and quantum field theory. MSDS extends known physics by a single small parameter.

2. Minimal Effective Model

2.1 Effective Lagrangian (λ/4 PRD Standard)

$$L = (1/2)\, \partial_\mu \text{phi}\, \partial^\mu \text{phi} - (\lambda/4)(\text{phi}^2 - v^2\Lambda^2)^2 + (\text{xi_R}/2)\, \text{phi}^2 R$$

phi: canonically normalised scalar | xi_R: non-minimal Ricci coupling (renamed from xi to avoid confusion with coherence length xi_0)

Footnote: The non-minimal coupling is renamed xi_R throughout Series G to avoid confusion with the coherence length xi_0 = 0.27 ly.

2.2 The λ Factor-of-4 Notation Sync

$$m_phi^2 = 2\, \text{lambda}\, v^2\, \text{Lambda}^2\ [\lambda/4 \text{ normalization, this paper}]$$

Equals 8 lambda_D v^2 Lambda^2 from Paper D's convention ($V=\lambda_D(\phi^2-v^2\Lambda^2)^2$)

$$\text{lambda} = m_phi^2/(2\, v^2\, M_Pl^2) = (7.660e\text{-}23)^2/(2\times 0.00017\times(1.22e{+}19)^2) = 1.159e\text{-}97$$

= 4 × lambda_D = 4 × 2.899e-98 — physical mass identical across conventions ✓

NATURALNESS: lambda ~ 1.16×10■■■ is technically natural in the sense that setting lambda = 0 restores an exact shift symmetry (phi → phi + const). All radiative corrections are proportional to lambda itself — no fine-tuning runaway. UV completion must address the origin of this scale; this is an open problem.

Symbol	Value	Derivation	Status
v^2	0.00017	Fitted to 1.73 Mpc clustering scale	Free parameter
lambda (λ/4 conv.)	1.159e-97	$m_phi^2/(2v^2M_Pl^2)$	Derived
m_phi	7.660e-23 eV	From v^2, lambda, M_Pl	Derived
f_c	18.52 nHz	$m_phi\ c^2/h$	Derived
xi_0	0.2722 ly	hbar/(m_phi c)	Derived
tau_lab	1.23e-04 s	tau_cosmic × L/xi_0	Derived

Figure 1: Series G summary. Top row — six phenomena mapped to $v^2 = 0.00017$ with tier labels; EFT parameters (λ/4 notation, xi_R, limiting case $v^2 \to 0$); Casimir torque $T(\theta) = T_■ v^2 \sin(2\theta)$ showing 170 ppm modulation. Bottom row — sonoluminescence harmonics (n=5: 629 nm orange-red visible); falsification tiers A (cosmological kill) and B (laboratory constrain); all lab scaling relations verified numerically.

3. Six Phenomena: Core Predictions

3.1 Casimir Torque Anisotropy — Laboratory Smoking Gun

The torsion offset v^2 induces chirality in the vacuum stress tensor. Between two parallel anisotropic plates (birefringent coating or patterned surface), the lowest-order anisotropic invariant yields:

$$T(\theta) = T_{\blacksquare} \times v^2 \times \sin(2\theta)$$

Directional modulation amplitude: $v^2 = 0.00017 = 170$ ppm | $T_{\blacksquare}$ calibrated by experiment

A relative modulation of $v^2 = 0.00017 = 170$ ppm is within reach of next-generation Casimir experiments with rotating anisotropic plates. Full derivation of the geometric factor: companion paper [Cas]. Tier 2 (testable now).

3.2 Quantum Eraser — Lattice Shear Memory

The vacuum lattice is viscoelastic. A particle traversing the lattice creates a shear wake whose lifetime is set by the relaxation time tau = 1/(gamma × f_c) with gamma ~ $v^2 = 0.00017$:

$$\text{tau_cosmic} = 1/(\text{gamma} \times \text{f_c}) = 3.176\text{e+11 s}$$

tau_lab = tau_cosmic × (L_lab/xi_0) = 1.23e-04 s for L_lab = 1 m

'Delayed choice' reads the stored wake before the lattice relaxes. This introduces no hidden variables and does not modify quantum probabilities. Testable: the effect should vanish for measurement intervals >> tau_lab = $1.2\times10^{\blacksquare\blacksquare}$ s. Tier 2. Full derivation: companion paper [QEr].

3.3 Sonoluminescence — Localised Vacuum Mode Coupling

$$\text{f_lab} = \text{f_c} \times (\text{xi_0/R}) \times \text{n}$$

R = 0.5 μm (bubble at collapse) | xi_0/R = 5.151e+21

$$\text{n=5: f = 477.1 THz} \rightarrow \lambda = 629 \text{ nm (visible orange-red)}$$

Testable: harmonic series in the spectrum; n=5 falls in visible range

> *COLOR CORRECTION (Omega): The paper originally stated 'blue/green.' 477 THz = 629 nm is orange-red, not blue-green. Blue-green requires 526-667 THz (n=6-7 in this model). The physical prediction — visible light from vacuum mode coupling — remains valid. The color is corrected here. Full derivation: companion paper [Sonol].*

3.4 Cosmic Mpemba Effect — Little Red Dots

High-energy regions reach the critical temperature earlier and crystallise faster. This explains JWST Little Red Dots: over-massive galaxies at z~5-8 that nucleated in the highest-energy lattice nodes. Prediction: the 1.73 Mpc clustering amplitude should be stronger in higher-energy (higher-z) regions. Tier 1 — tentative support from Paper E (global 1.20σ, p=0.231). Not a detection.

3.5 Dark Matter — Superfluid Vortex Lattice

The flat rotation velocity v_flat ~ 200 km/s gives a 2D vortex density:

n = m_phi v_flat²/hbar = 5.170e-14 m■² → d = 1/sqrt(n) = 4398 km

2D vortex spacing — CORRECTED from 14,000 km in submitted draft

DM SPACING CORRECTION (Omega): The submitted draft stated d ≈ 14,000 km. The correct value from d = 1/sqrt(n) is 4398 km. The n = 5.2×10■¹■ m■² in the draft is correct; the d calculation was off by factor ~3. In a realistic 3D galactic halo, the effective spacing requires a full Gross-Pitaevskii simulation with vertical stacking over ~10 kpc thickness. We leave this to companion paper [DM]. The 2D result itself is Tier 2 (effective model).

3.6 RDAC Sidereal Modulation — Laboratory Test

Delta-I/I ~ v² × S = 85 ppm (S = geometric suppression factor)

Period: 23.93 hr (sidereal day) | Tier 2 — fully model-independent

4. Null Hypothesis and No-Go

To assess whether the 1.73 Mpc excess could arise by chance in ΛCDM, 10^5 mock realisations of the JWST footprint were generated (Paper E). A peak of equal or greater amplitude occurred in 23,100 of 100,000 runs: p = 0.231, Z_global = 1.20σ (two-tailed). This feature is not strong enough to claim a detection. It is a tentative seed worth further investigation.

No-go theorem: a Lorentz-invariant vacuum without additional structure cannot produce a directional Casimir anisotropy without explicit boundary-induced symmetry breaking. Detection of such anisotropy requires an extension of the standard model.

5. Unified Phenomenon Table

Phenomenon	Anchor	Prediction	Tier	Companion
Casimir torque	v^2	170 ppm sin(2θ)	2 (testable)	[Cas]
Quantum eraser	v^2, f_c	tau_lab = 1.2×10■■ s	2 (testable)	[QEr]
Sonoluminescence	f_c, xi_0	n=5 → 629 nm visible	2 (testable)	[Sonol]
LRDs (1.73 Mpc)	v^2	1.2σ seed (Paper E)	1 (tentative)	[Mpemba]
Dark matter	m_phi	Vortex d~4398 km (2D)	2 (model)	[DM]
RDAC modulation	v^2	85 ppm at 23.93 hr	2 (proposed)	[RDAC]

Table 1: All six phenomena anchored to v^2 = 0.00017 and derived scales. Tier 1 requires independent confirmation; Tier 2 requires dedicated experiments.

6. Falsification Tiers

Tier A — Cosmological (kills the core framework):

- 1.73 Mpc clustering NOT confirmed by Roman Space Telescope (2028-29)
- 18.52 nHz GWB feature absent in SKA-era PTA data at 5σ

Tier B — Laboratory (constrains the model):

- Casimir torque shows no anisotropy at v^2 level with rotating anisotropic plates
- Sonoluminescence shows no harmonic structure scaling with bubble size
- Quantum eraser effect persists at measurement intervals >> tau_lab
- RDAC shows no sidereal modulation down to 10■■ relative amplitude

Failure of ANY Tier A test falsifies the core framework independently of Tier B results. Tier B results constrain or refine the model but do not independently falsify it.

7. Brief Comparison with String Theory

String theory requires extra dimensions compactified at sub-Planck scales, inaccessible to current experiment. MSDS works entirely within 3+1 spacetime — the 'extra' structure is the phase state of the vacuum, not additional dimensions. This is ontological parsimony: one testable constant (v^2) plus a technically natural coupling (λ) explains a range of phenomena using only the dimensions we observe. MSDS does not claim string theory is wrong; it offers a complementary, testable approach focused on low-energy vacuum structure.

SERIES G — PHENOMENOLOGICAL VACUUM-LATTICE FRAMEWORK

Single parameter: $v^2 = 0.00017$ (limit $v^2 \to 0$ = standard ΛCDM + QFT)

λ = 1.159e-97 (λ/4 PRD notation; = 4×λ_D from Paper D)

Six phenomena anchored to f_c=18.52 nHz, xi_0=0.2722 ly

Casimir torque: 170 ppm sin(2θ) | QE: tau_lab=1.23e-04 s

Sono: n=5 → 629 nm (visible) | DM: vortex d~4398 km (2D, corrected)

RDAC: 85 ppm at 23.93 hr | LRDs: 1.2σ seed (Roman will decide)

Tier A kills the framework: Roman clustering + PTA resonance

Tier B constrains: four lab experiments

Appendix C quarantined: warp-drive, metamaterials, tachyon (inspirational only)

APPENDIX C — SPECULATIVE LONG-TERM OUTLOOK | WARNING: INSPIRATIONAL ONLY — NOT QUANTITATIVE PREDICTIONS — NOT REQUIRED FOR CORE FRAMEWORK

The following ideas are highly speculative and presented solely to illustrate potential long-term reach of the vacuum-lattice picture. They are not quantitative predictions, are not required for the core framework, and should not be interpreted as claims of the model.

C.1 Torsional Solitons (Warp-Drive Analogues)

High-energy phase-shifting could create a stable 'soliton bubble' — a self-sustaining distortion of the torsion field moving at $v_{shear} = c(1+v^2)$, while signal velocity remains $\leq c$. Analogous to Alcubierre drive but without exotic matter. No FTL signaling implied.

C.2 Vacuum-Resonant Metamaterials

If a material's atomic lattice is phase-locked to the vacuum lattice ($a_{crystal} = \xi_0/N$ for integer N), the material could be vacuum-reinforced. Speculative pathway toward novel properties via v^2 stiffness amplification.

C.3 Tachyonic Phase Waves (Torsional Shear Modes)

The torsion offset modifies the vacuum shear modulus, giving torsional shear waves a phase velocity $v_{shear} = c(1+v^2)$. Such phase-slip waves carry no information and do not violate causality. Not particles — a new pathway for understanding anomalous dispersion.

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